3.1.3 Automated Warfare

AIs speed up the pace of war, which makes AIs more necessary. AIs can quickly process a large amount of data, analyze complex situations, and provide helpful insights to commanders. With ubiquitous sensors and advanced technology on the battlefield, there is tremendous incoming information. AIs help make sense of this information, spotting important patterns and relationships that humans might miss. As these trends continue, it will become increasingly difficult for humans to make well-informed decisions as quickly as necessary to keep pace with AIs. This would further pressure militaries to hand over decisive control to AIs. The continuous integration of AIs into all aspects of warfare will cause the pace of combat to become faster and faster. Eventually, we may arrive at a point where humans are no longer capable of assessing the ever-changing battlefield situation and must cede decision-making power to advanced AIs.

Automatic retaliation can escalate accidents into war. There is already willingness to let computer systems retaliate automatically. In 2014, a leak revealed to the public that the NSA was developing a system called MonsterMind, which would autonomously detect and block cyberattacks on US infrastructure [42]. It was suggested that in the future, MonsterMind could automatically initiate a retaliatory cyberattack with no human involvement. If multiple combatants have policies of automatic retaliation, an accident or false alarm could quickly escalate to full-scale war before humans intervene. This would be especially dangerous if the superior information processing capabilities of modern AI systems makes it more appealing for actors to automate decisions regarding nuclear launches.

History shows the danger of automated retaliation. On September 26, 1983, Stanislav Petrov, a lieutenant colonel of the Soviet Air Defense Forces, was on duty at the Serpukhov-15 bunker near Moscow, monitoring the Soviet Union's early warning system for incoming ballistic missiles. The system indicated that the US had launched multiple nuclear missiles toward the Soviet Union. The protocol at the time dictated that such an event should be considered a legitimate attack, and the Soviet Union would respond with a nuclear counterstrike. If Petrov had passed on the warning to his superiors, this would have been the likely outcome. Instead, however, he judged it to be a false alarm and ignored it. It was soon confirmed that the warning had been caused by a rare technical malfunction. If an AI had been in control, the false alarm could have triggered a nuclear war.

Figure 8: A military AI arms race could pressure countries into delegating many crucial decisions over armaments to AIs. Integrating AIs into nuclear command and control could heighten the risk of global catastrophe as the potential for accidents and increased pace of war may lead to unintended escalations and confrontations.

AI-controlled weapons systems could lead to a flash war. Autonomous systems are not infallible. We have already witnessed how quickly an error in an automated system can escalate in the economy. Most notably, in the 2010 Flash Crash, a feedback loop between automated trading algorithms amplified ordinary market fluctuations into a financial catastrophe in which a trillion dollars of stock value vanished in minutes [43]. If multiple nations were to use AIs to automate their defense systems, an error could be catastrophic, triggering a spiral of attacks and counter-attacks that would happen too quickly for humans to step in—a flash war. The market quickly recovered from the 2010 Flash Crash, but the harm caused by a flash war could be catastrophic.

Automated warfare could reduce accountability for military leaders. Military leaders may at times gain an advantage on the battlefield if they are willing to ignore the laws of war. For example, soldiers may be able

to mount stronger attacks if they do not take steps to minimize civilian casualties. An important deterrent to this behavior is the risk that military leaders could eventually be held accountable or even prosecuted for war crimes. Automated warfare could reduce this deterrence effect by making it easier for military leaders to escape accountability by blaming violations on failures in their automated systems.

AIs could make war more uncertain, increasing the risk of conflict. Although states that are already wealthier and more powerful often have more resources to invest in new military technologies, they are not necessarily always the most successful at adopting them. Other factors also play an important role, such as how agile and adaptive a military can be in incorporating new technologies [44]. Major new weapons innovations can therefore offer an opportunity for existing superpowers to bolster their dominance, but also for less powerful states to quickly increase their power by getting ahead in an emerging and important sphere. This can create significant uncertainty around if and how the balance of power is shifting, potentially leading states to incorrectly believe they could gain something from going to war. Even aside from considerations regarding the balance of power, rapidly evolving automated warfare would be unprecedented, making it difficult for actors to evaluate their chances of victory in any particular conflict. This would increase the risk of miscalculation, making war more more likely.

3.1.4 Actors May Risk Extinction Over Individual Defeat

"I know not with what weapons World War III will be fought, but World War IV will be fought with sticks and stones." *Einstein*

Competitive pressures make actors more willing to accept the risk of extinction. During the Cold War, neither side desired the dangerous situation they found themselves in. There were widespread fears that nuclear weapons could be powerful enough to wipe out a large fraction of humanity, potentially even causing extinction—a catastrophic result for both sides. Yet the intense rivalry and geopolitical tensions between the two superpowers fueled a dangerous cycle of arms buildup. Each side perceived the other's nuclear arsenal as a threat to its very survival, leading to a desire for parity and deterrence. The competitive pressures pushed both countries to continually develop and deploy more advanced and destructive nuclear weapons systems, driven by the fear of being at a strategic disadvantage. During the Cuban Missile Crisis, this led to the brink of nuclear war. Even though the story of Arkhipov preventing the launch of a nuclear torpedo wasn't declassified until decades after the incident, President John F. Kennedy reportedly estimated that he thought the odds of nuclear war beginning during that time were "somewhere between one out of three and even." This chilling admission highlights how the competitive pressures between militaries have the potential to cause global catastrophes.

Individually rational decisions can be collectively catastrophic. Nations locked in competition might make decisions that advance their own interests by putting the rest of the world at stake. Scenarios of this kind are collective action problems, where decisions may be rational on an individual level yet disastrous for the larger group [45]. For example, corporations and individuals may weigh their own profits and convenience over the negative impacts of the emissions they create, even if those emissions collectively result in climate change. The same principle can be extended to military strategy and defense systems. Military leaders might estimate, for instance, that increasing the autonomy of weapon systems would mean a 10 percent chance of losing control over weaponized superhuman AIs. Alternatively, they might estimate that using AIs to automate bioweapons research could lead to a 10 percent chance of leaking a deadly pathogen. Both of these scenarios could lead to catastrophe or even extinction. The leaders may, however, also calculate that refraining from these developments will mean a 99 percent chance of losing a war against an opponent. Since conflicts are often viewed as existential struggles by those fighting them, rational actors may accept an otherwise unthinkable 10 percent chance of human extinction over a 99 percent chance of losing a war. Regardless of the particular nature of the risks posed by advanced AIs, these dynamics could push us to the brink of global catastrophe.

Technological superiority does not guarantee national security. It is tempting to think that the best way of guarding against enemy attacks is to improve one's own military prowess. However, in the midst of competitive

pressures, all parties will tend to advance their weaponry, such that no one gains much of an advantage, but all are left at greater risk. As Richard Danzig, former Secretary of the Navy, has observed, "The introduction of complex, opaque, novel, and interactive technologies will produce accidents, emergent effects, and sabotage. On a number of occasions and in a number of ways, the American national security establishment will lose control of what it creates... deterrence is a strategy for reducing attacks, not accidents" [46].

Cooperation is paramount to reducing risk. As discussed above, an AI arms race can lead us down a hazardous path, despite this being in no country's best interest. It is important to remember that we are all on the same side when it comes to existential risks, and working together to prevent them is a collective necessity. A destructive AI arms race benefits nobody, so all actors would be rational to take steps to cooperate with one another to prevent the riskiest applications of militarized AIs. As Dwight D. Eisenhower reminded us, "The only way to win World War III is to prevent it."

We have considered how competitive pressures could lead to the increasing automation of conflict, even if decision-makers are aware of the existential threat that this path entails. We have also discussed cooperation as being the key to counteracting and overcoming this collective action problem. We will now illustrate a hypothetical path to disaster that could result from an AI arms race.

Story: Automated Warfare

As AI systems become increasingly sophisticated, militaries start involving them in decision-making processes. Officials give them military intelligence about opponents' arms and strategies, for example, and ask them to calculate the most promising plan of action. It soon becomes apparent that AIs are reliably reaching better decisions than humans, so it seems sensible to give them more influence. At the same time, international tensions are rising, increasing the threat of war.

A new military technology has recently been developed that could make international attacks swifter and stealthier, giving targets less time to respond. Since military officials feel their response processes take too long, they fear that they could be vulnerable to a surprise attack capable of inflicting decisive damage before they would have any chance to retaliate. Since AIs can process information and make decisions much more quickly than humans, military leaders reluctantly hand them increasing amounts of retaliatory control, reasoning that failing to do so would leave them open to attack from adversaries.

While for years military leaders had stressed the importance of keeping a "human in the loop" for major decisions, human control is nonetheless gradually phased out in the interests of national security. Military leaders understand that their decisions lead to the possibility of inadvertent escalation caused by system malfunctions, and would prefer a world where all countries automated less; but they do not trust that their adversaries will refrain from automation. Over time, more and more of the chain of command is automated on all sides.

One day, a single system malfunctions, detecting an enemy attack when there is none. The system is empowered to launch an instant "retaliatory" attack, and it does so in the blink of an eye. The attack causes automated retaliation from the other side, and so on. Before long, the situation is spiraling out of control, with waves of automated attack and retaliation. Although humans have made mistakes leading to escalation in the past, this escalation between mostly-automated militaries happens far more quickly than any before. The humans who are responding to the situation find it difficult to diagnose the source of the problem, as the AI systems are not transparent. By the time they even realize how the conflict started, it is already over, with devastating consequences for both sides.

3.2 Corporate AI Race

Competitive pressures exist in the economy, as well as in military settings. Although competition between companies can be beneficial, creating more useful products for consumers, there are also pitfalls. First, the